20.11.25

OPENCAMPUS.sh

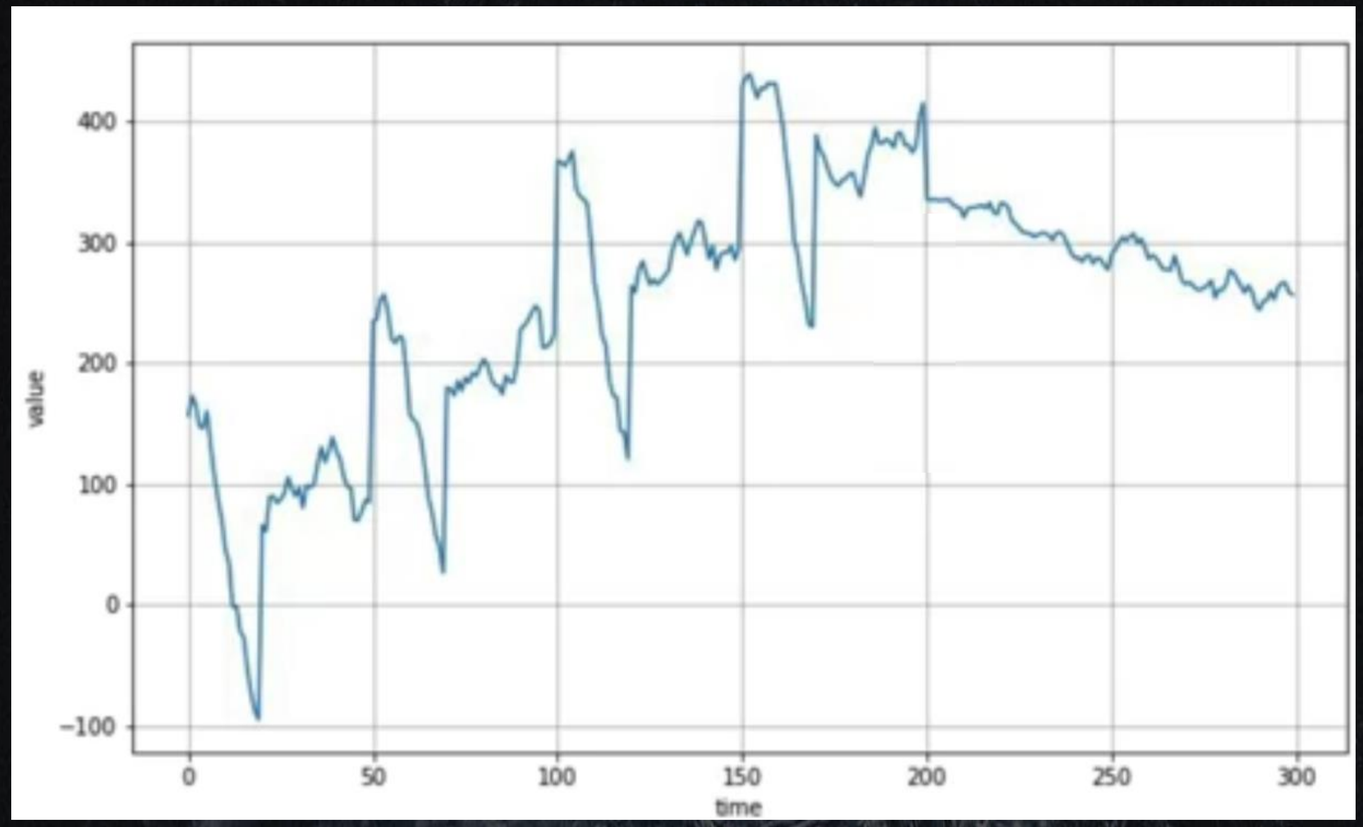
Introduction to Data Science and Machine Learning

TIME SERIES ANALYSIS AND INTRODUCTION INTO MACHINE LEARNING

- Different patterns in time series
- Procedure for training machine learning algorithms
- Testing your model on Kaggle
- Current developments in AI

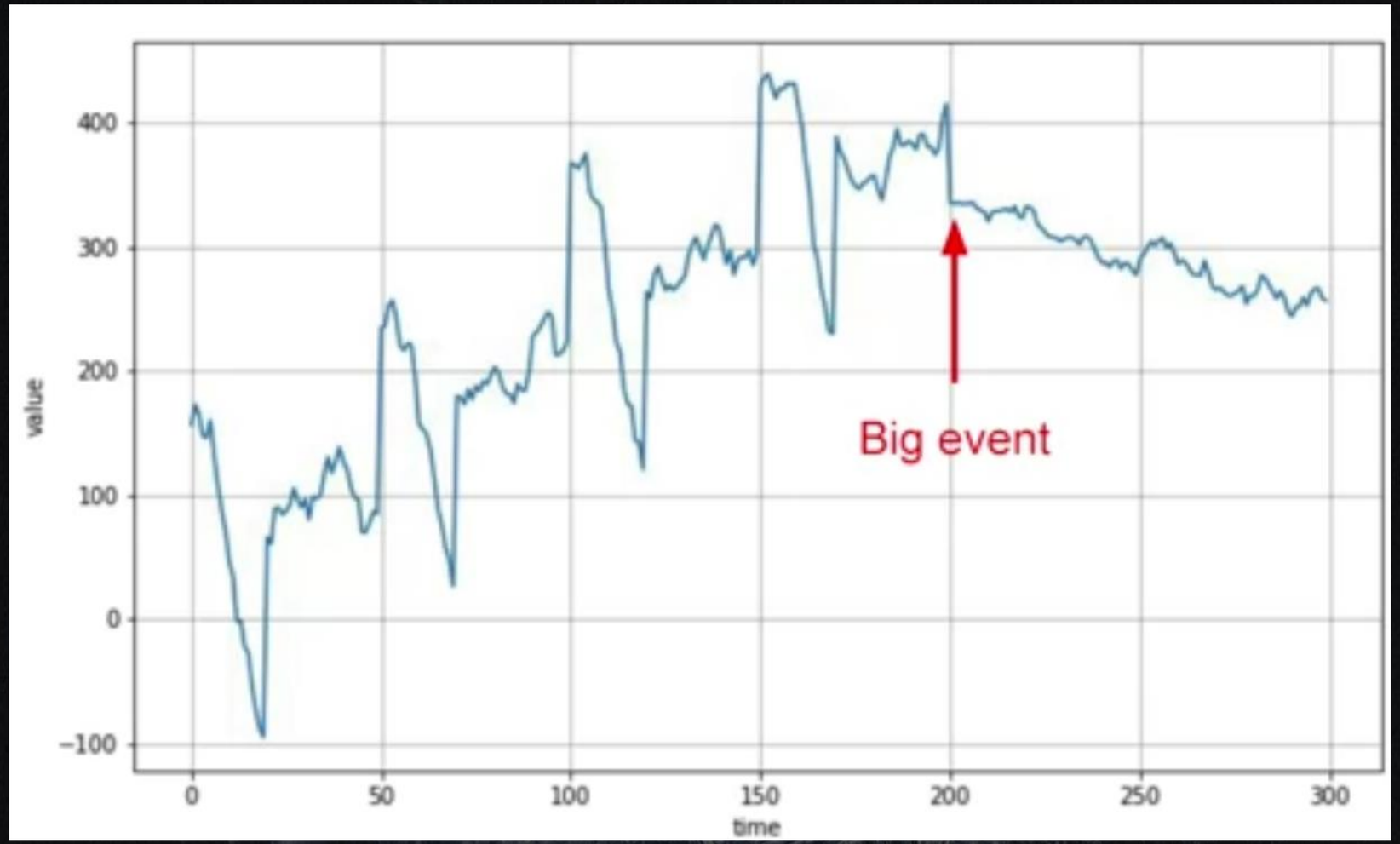
DISCUSSION

What different types of patterns can be identified in the displayed figure?



DeepLearning.AI. (o. J.). Common patterns in time series—Sequences and Prediction. Abgerufen 7. Januar 2021, von Coursera website: https://www.coursera.org/lecture/tensorflow-sequences-time-series-and-prediction/common-patterns-in-time-series-dWnOy

NON-STATIONARITY

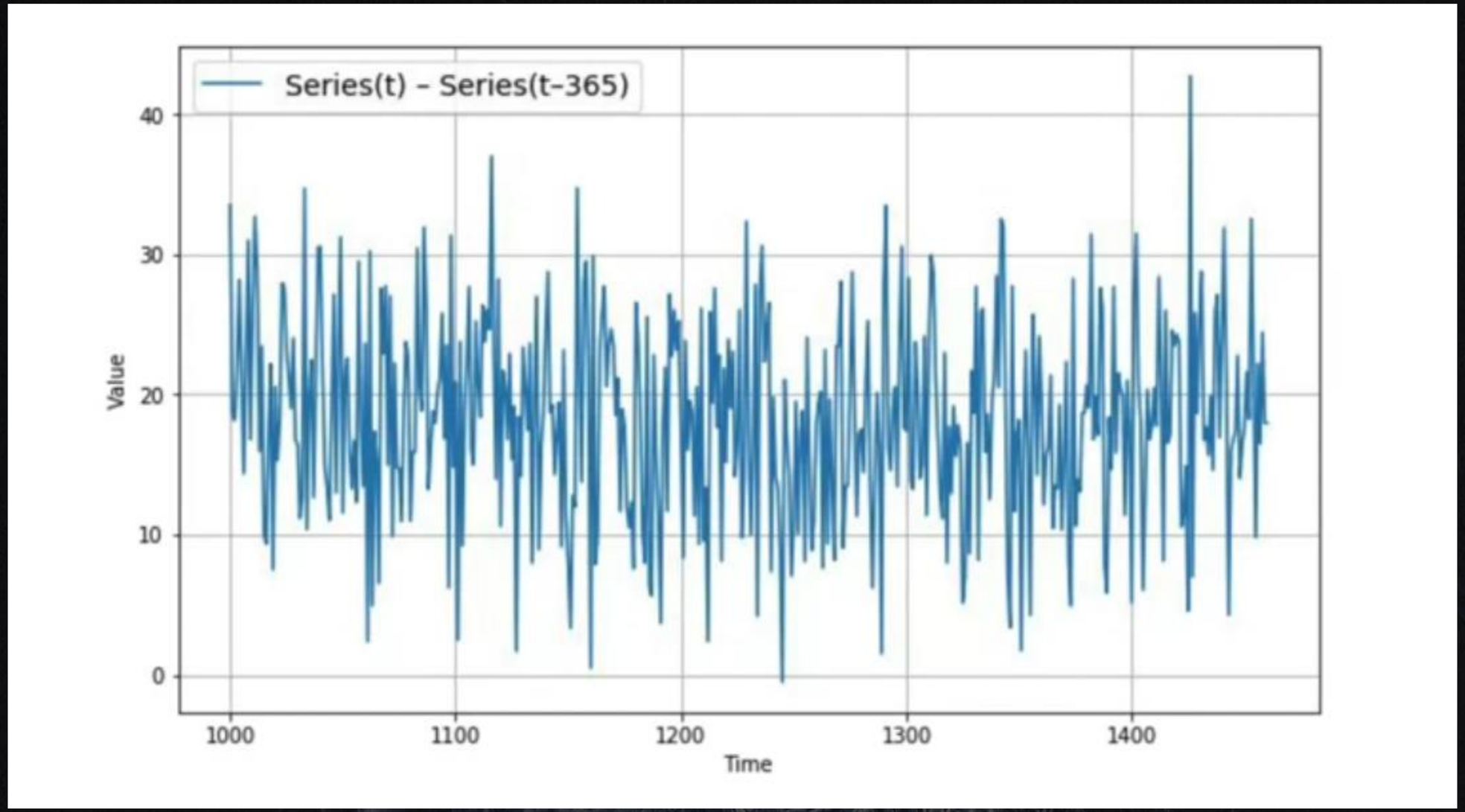


DeepLearning.AI. (o. J.). Common patterns in time series—Sequences and Prediction. Abgerufen 7. Januar 2021, von Coursera website: https://www.coursera.org/lecture/tensorflow-sequences-time-series-and-prediction/common-patterns-in-time-series-dWnOy

PATTERNS IN TIME SERIES

- Trends
- Seasonality / Cyclicality
- Noise
- Autocorrelation (serial correlation)

DIFFERENCING



DIFFERENCING

- Subtracting the previous observation from the current observation
- A method for removing trends, seasonality, or other potential autocorrelations in time series
- Helps to reduce autocorrelation and obtain stationary time series

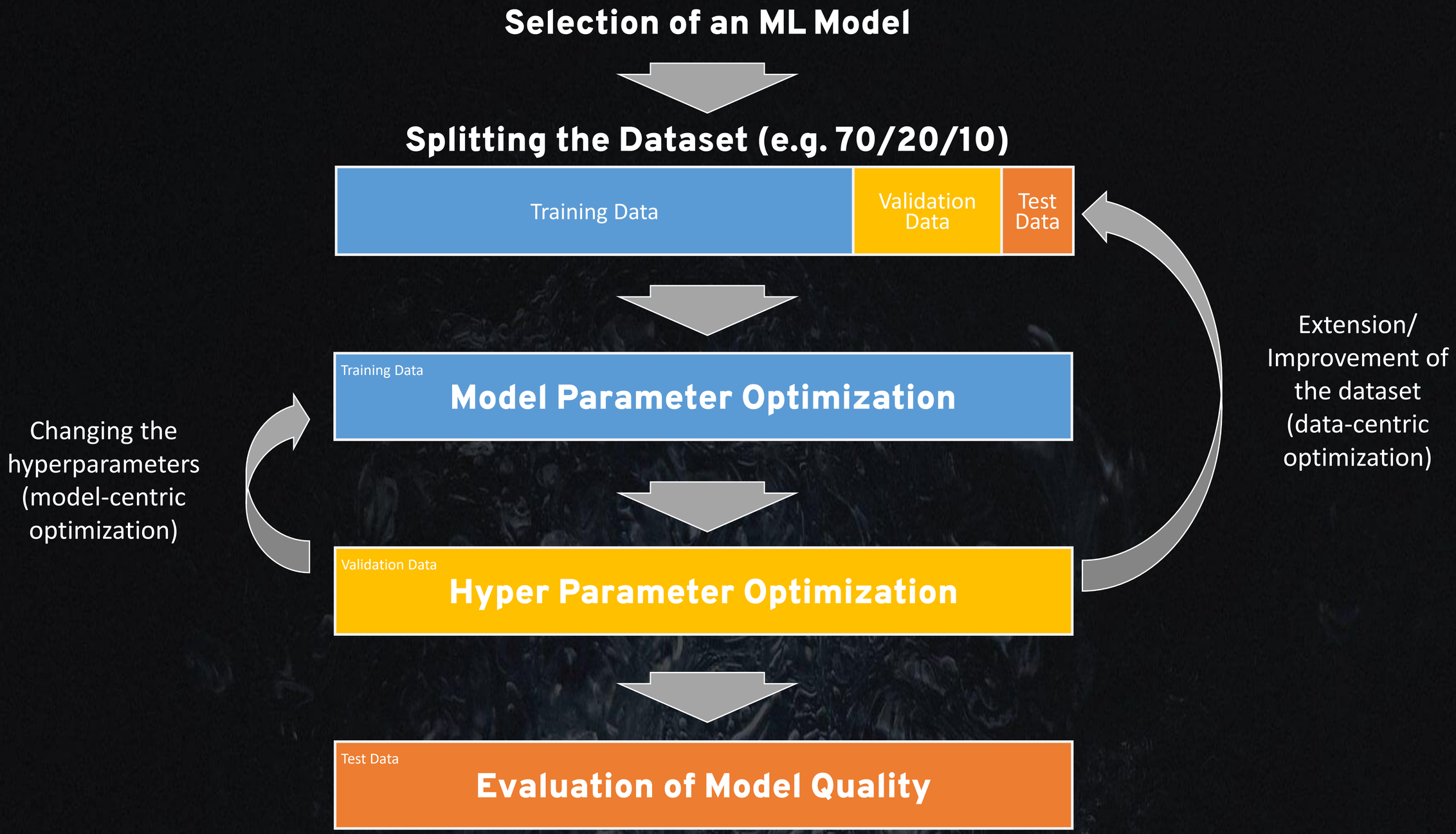
„LAG“

- The difference between consecutive observations is called the Lag-1 difference.
- More generally, a Lag-k difference refers to the difference between an observation and the observation k time steps earlier.
- The lag value should be chosen based on the temporal structure of the data.
- For time series with a seasonal component, the lag is typically set to match the length of the seasonal cycle (e.g., lag = 12 for monthly data with yearly seasonality).

MODELS FOR MACHINE LEARNING

Covered in this course:

- **Linear model**
- **Neural network**
- **(Support vector machine)**


Selection of an ML Model
Splitting the Dataset (e.g. 70/20/10)
Training Data
Validation Data
Test Data
Training Data
Model Parameter Optimization
Validation Data
Hyper Parameter Optimization
Test Data
Evaluation of Model Quality
Changing the hyperparameters (model-centric optimization)
Extension/ Improvement of the dataset (data-centric optimization)

MODEL PARAMETERS VS. HYPER PARAMETERS

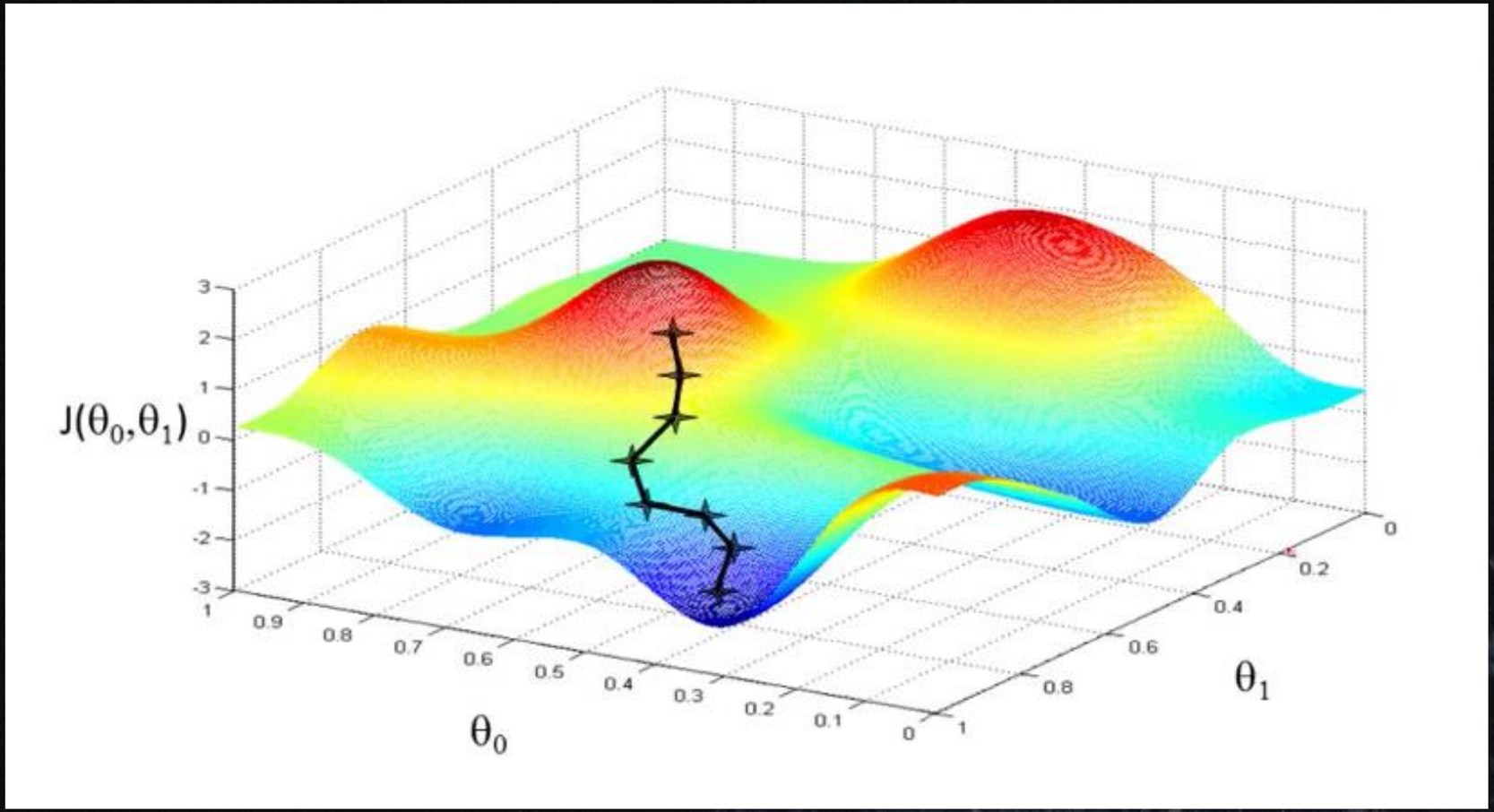
Model Parameters

- **Parameters that are optimized during training (weights and biases).**

Hyper Parameters

- **Parameters set before training (e.g., on how the parameters are optimized, how many are optimized, or how they relate to each other).**

OPTIMIZATION FUNCTION (OPTIMIZER)



- **Optimizes the model parameters only!**
- **Iterative method (Gradient Descent) to find the minimum of the cost function.**
- **The learning rate ("learning parameter") describes the step size for approaching the minimum.**

Quelle: https://www.coursera.org/learn/machine-learning

OPTIMIZATION OF THE MODEL PARAMETERS

Forward Propagation

- **Computation of predictions for a dataset using the current model parameters (weights and biases)**
- **Computation of the total cost or loss based on the difference between predictions and actual values**

Backward Propagation

- **Calculation of the contribution of each model parameter to the total loss**
- **Computation of gradients (i.e., partial derivatives of the loss with respect to parameter) to determine how to update the parameters to reduce the loss**

Gradient Descent (Optimization Step)

- **Use the derivates to update ethe parameters**
- **Typical update rule:**

```
New Value = Old Value – Learning Rate × Partial Derivative
```

FEATURES, LABELS, PARAMETER

Machine Learning	Statistics	Description
Feature, Input Variable	Independent Variable	Input data used for prediction
Label, Target Variable, Output	Dependent Variable	Observed outcomes used to train the model
Weights, (Modell) Parameter	Coefficients	Are optimized ("learned") during training

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Community
Created by fellow Kagglers

Playgr
Fun pra
problem

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These competitions are perfect for newcomers.



Titanic - Machine Learning from Disaster

Start here! Predict survival on the Ti...

Getting Started

15573 Teams

Knowledge Ongoing



House Prices - Advanced Regression...

Predict sales prices and practice fea...

Getting Started

4911 Teams

Knowledge Ongoing



Spaceship Titanic

Predict which passengers are transp...

Getting Started

2555 Teams

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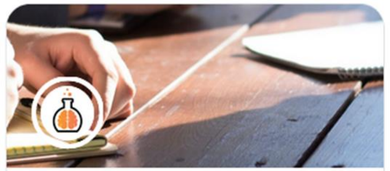
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LLM - Detect AI Generated Text

Identify which essay was written by ...

Featured · Code Competition

1740 Teams

$110,000 2 months to go



Open Problems – Single-Cell...

Predict how small molecules change...

Featured

1120 Teams

$100,000 3 days to go



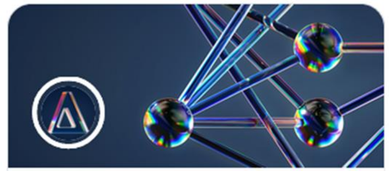
Stanford Ribonanza RNA Folding

Create a model that predicts the str...

Research

676 Teams

$100,000 10 days to go



Optiver - Trading at the Close

Predict US stocks closing movements

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$100,000 23 days to go

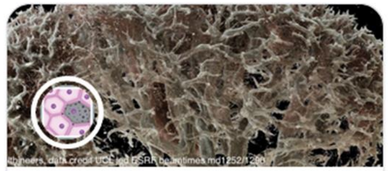


NFL Big Data Bowl 2024

Help evaluate tackling tactics and st...

Analytics

$100,000 a month to go



SenNet + HOA - Hacking the Human...

Segment vasculature in 3D scans of ...

Research · Code Competition

149 Teams

$80,000 2 months to go



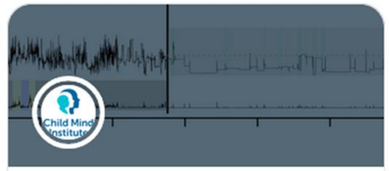
Linking Writing Processes to Writing...

Use typing behavior to predict essa...

Featured · Code Competition

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Child Mind Institute - Detect Sleep States

Detect sleep onset and wake from w...

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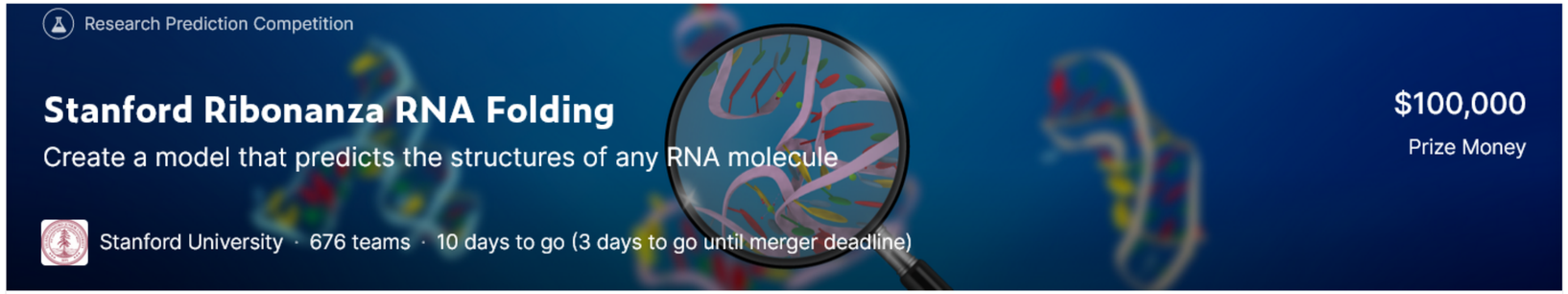
Code

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Overview | Data | Code | Models | Discussion | Leaderboard | Rules

Join Competition

Leaderboard

Raw Data | Refresh

Search leaderboard

Public | Private

This leaderboard is calculated with approximately 20% of the test data. The final results will be based on the other 80%, so the final standings may be different.

Prize Contenders

#	Team	Members	Score	Entries	Last	Join
1	HandFold		0.13733	85	40m	
2	DI		0.13773	108	3h	

SPLITTING A DATASET WITHOUT TIME SERIES (!)

1) Shuffling the order (randomization)

2) Definition of the sizes of training, validation, and test datasets in percentages

3) Splitting the dataset

EXAMPLE OF DATASET SPLITTING (NO TIME SERIES)

```python
import pandas as pd
from sklearn.model_selection import train_test_split

# Load the dataset
url = "https://raw.githubusercontent.com/opencampus-sh/einfuehrung-in-data-science-und-ml/main/house_pricing_data/house_pricing_train.csv"
data = pd.read_csv(url)

# Set a random seed for reproducibility
random_state = 42

# First, split the data into training (70%) and remaining (30%)
train_data, remaining_data = train_test_split(data, train_size=0.7, random_state=random_state)

# Now split the remaining data into validation (2/3 of remaining) and test (1/3 of remaining)
validation_data, test_data = train_test_split(remaining_data, test_size=0.3333, random_state=random_state)

# Check the dimensions of the datasets
print("Training dataset dimensions:", train_data.shape)
print("Validation dataset dimensions:", validation_data.shape)
print("Test dataset dimensions:", test_data.shape)
```

SPLITTING A DATASET WITH TIME SERIES

1) Ordering the time series

2) Definition of time series windows for validation and test datasets

3) Splitting the dataset

EXAMPLE OF DATASET SPLITTING (TIME SERIES)

```
import pandas as pd

# Sample data
data = pd.DataFrame({
    'date': pd.date_range(start='2021-01-01', periods=100, freq='D'),
    'value': range(100)
})

# Ensure the data is sorted by date
data = data.sort_values(by='date')

print (data.head())

# Define your date thresholds
train_end_date = '2021-03-31'
validation_end_date = '2021-04-30'

# Convert to datetime if not already
data['date'] = pd.to_datetime(data['date'])

# Split the data based on the date thresholds
train_data = data[data['date'] <= train_end_date]
validation_data = data[(data['date'] > train_end_date) & (data['date'] <= validation_end_date)]
test_data = data[data['date'] > validation_end_date]

# Check the dimensions of the datasets
print("Training dataset dimensions:", train_data.shape)
print("Validation dataset dimensions:", validation_data.shape)
print("Test dataset dimensions:", test_data.shape)
```

BREAKOUT

- Download the test data set from Kaggle and merge it with your current data set.

- Split your dataset into a training dataset from 01.07.2013 to 31.07.2017, a validation dataset from 01.08.2017 to 31.07.2018, and the test set from 01.08.2018 to 31.07.2019.

  (Ensure that the number of rows and the defined IDs match those in the downloaded test dataset!)

LEARNING RESOURCES

- Complete the sections on „Linear Regression“ and „Multiple Linear Regression“ from Module 4 of the course Python for Data Analytics by Deeplearning.ai (about 2 hours).

TASKS

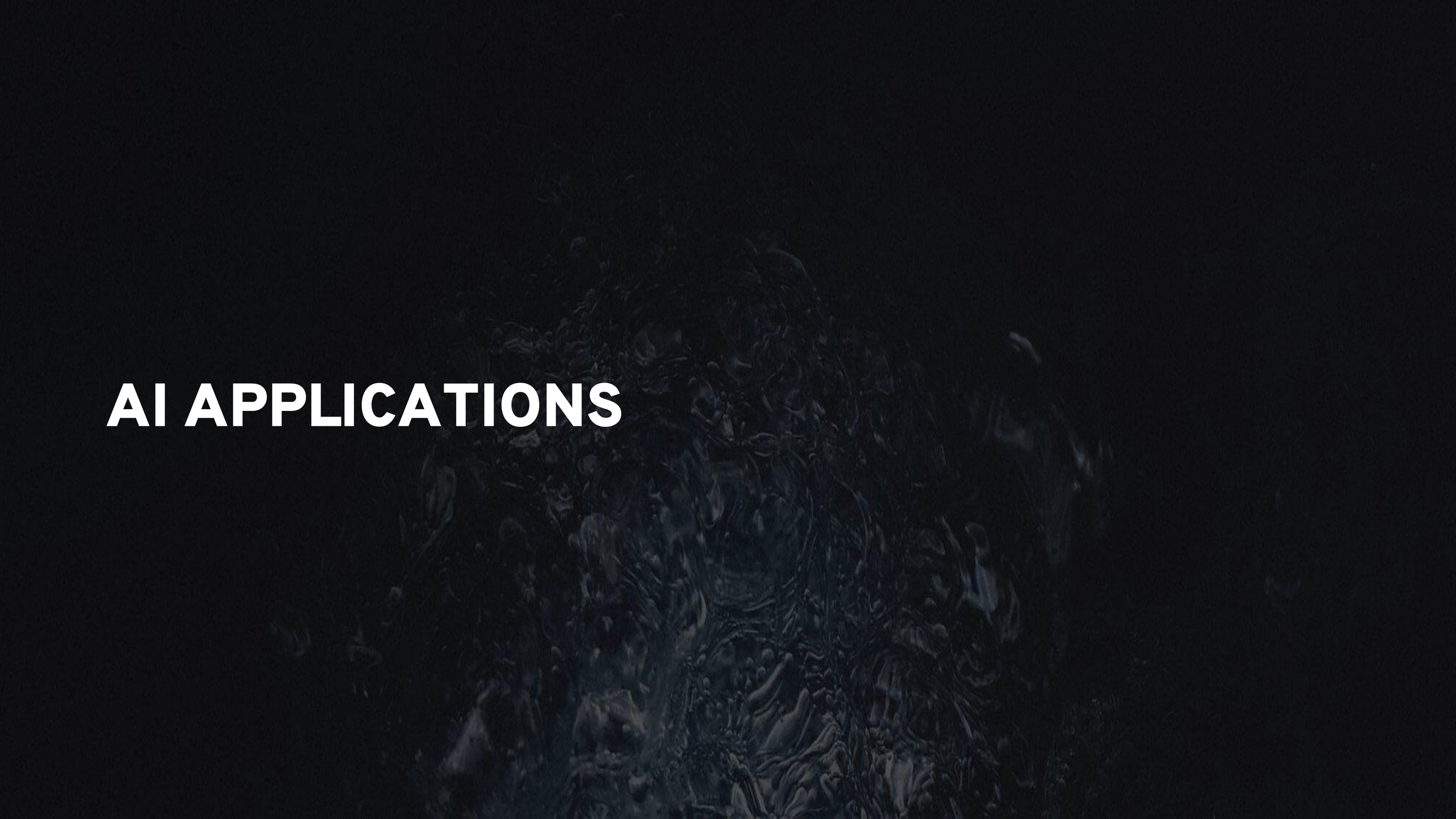
- Download the test data set from Kaggle and merge it with your current data set.

- Split your dataset into a training dataset from 01.07.2013 to 31.07.2017, a validation dataset from 01.08.2017 to 31.07.2018, and the test set from 01.08.2018 to 31.07.2019.

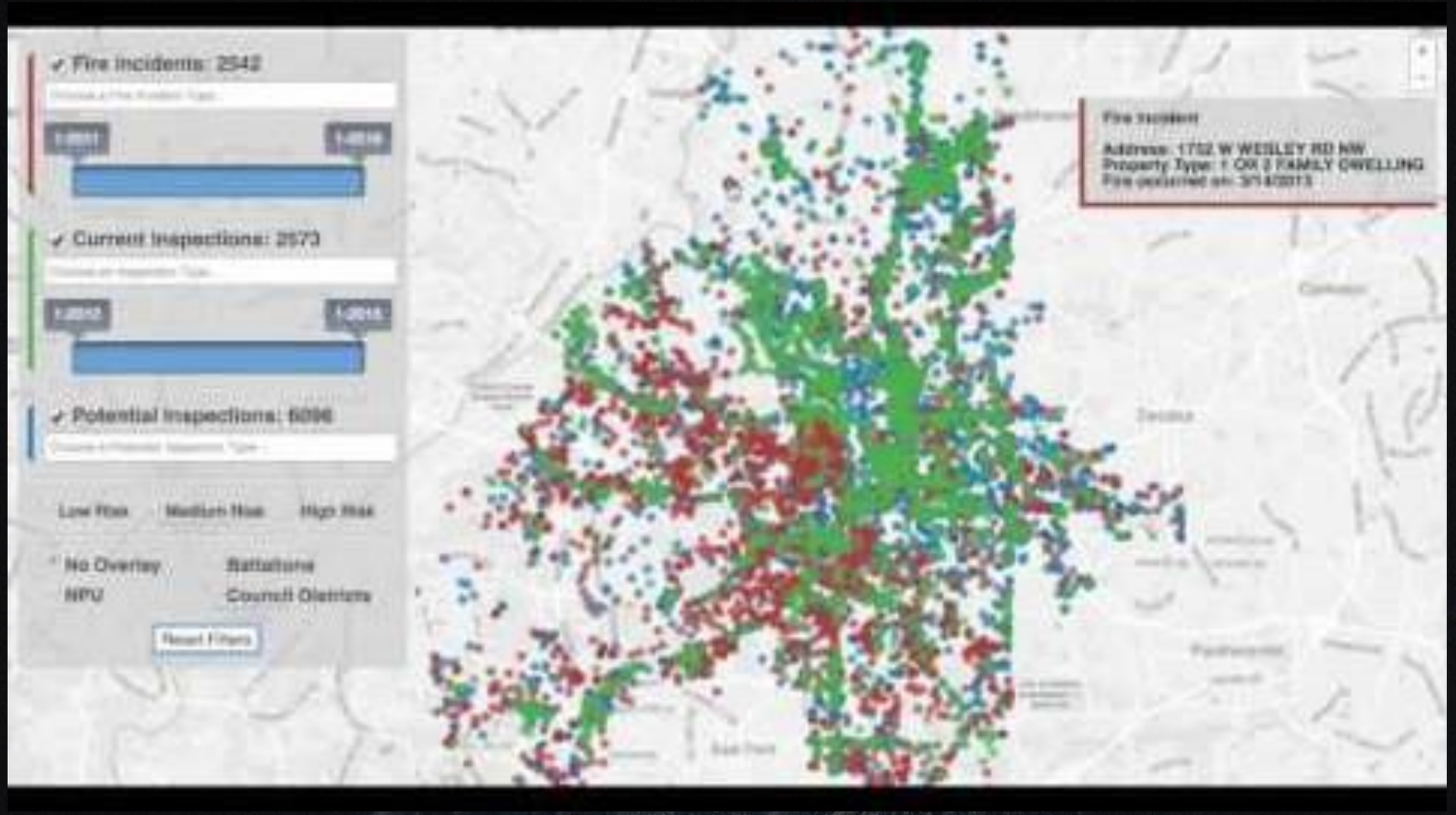
  (Ensure that the number of rows and the defined IDs match those in the downloaded test dataset!)

- Further enrich the dataset with additional variables that may be relevant for estimating the sales (based on the time series itself or from external sources).

AI APPLICATIONS



RISK FORECASTING

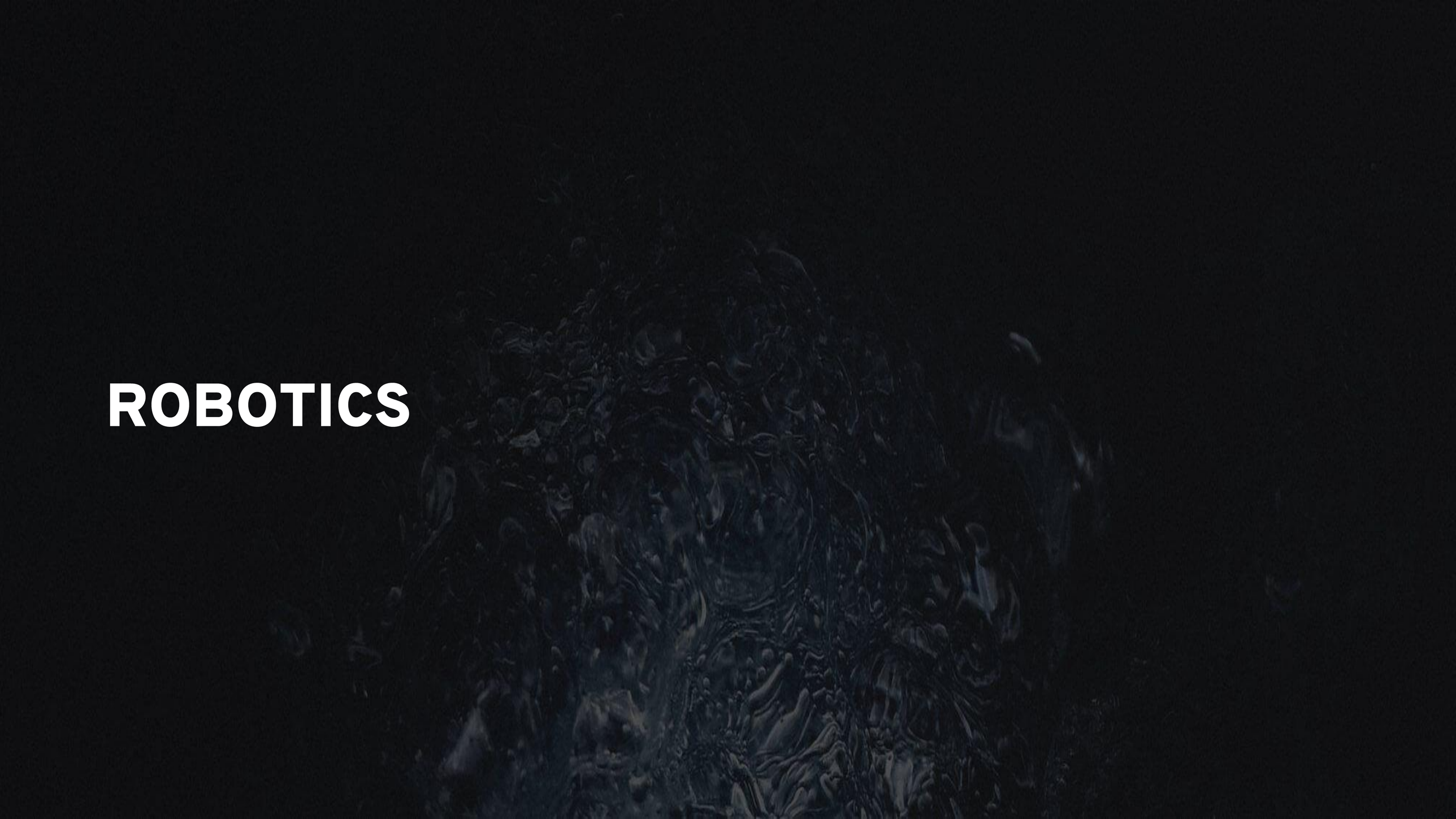


CHESS/ GAMING COMPUTER



AlphaGo – The Movie:
https://www.youtube.com/watch?v=WXuK6gekU1Y

ROBOTICS




Unitree A2
Stellar Hunter
UNITREE


UNITREE G1
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SOCIAL BOTS

- **Replika**
  for a personal, emotional friend
- **Character.ai**
  with a wide variety of characters
- **Dream Companion**
  for a ”highly customizable roleplaying experience”

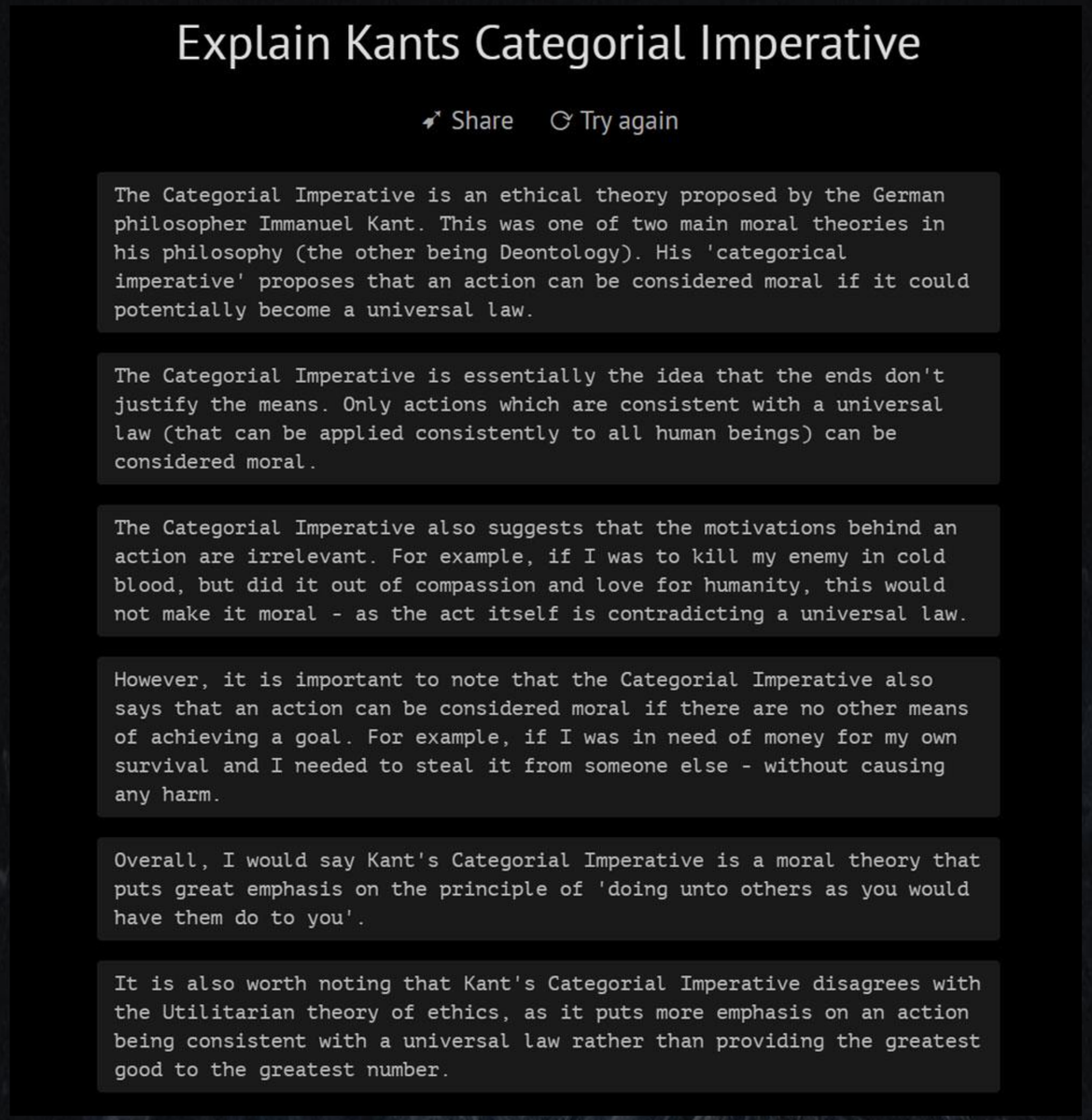


Advay Pal [@advaypal]. (2023, April 26). what have we done? Https://t.co/DVC0zL2Xgh [Tweet]. Retrieved October 9, 2023, from Twitter website: https://twitter.com/advaypal/status/1651355572980310016

NATURAL LANGUAGE PROCESSING



GPT-3



Schreiner, M. (2020, August 31). GPT-3: Diese fünf Apps zeigen das KI-Potenzial [Blog]. Retrieved May 4, 2021, from MIXED website: https://mixed.de/gpt-3-diese-fuenf-apps-zeigen-das-ki-potenzial/

MULTIMODAL MODELS





CREATIVITY

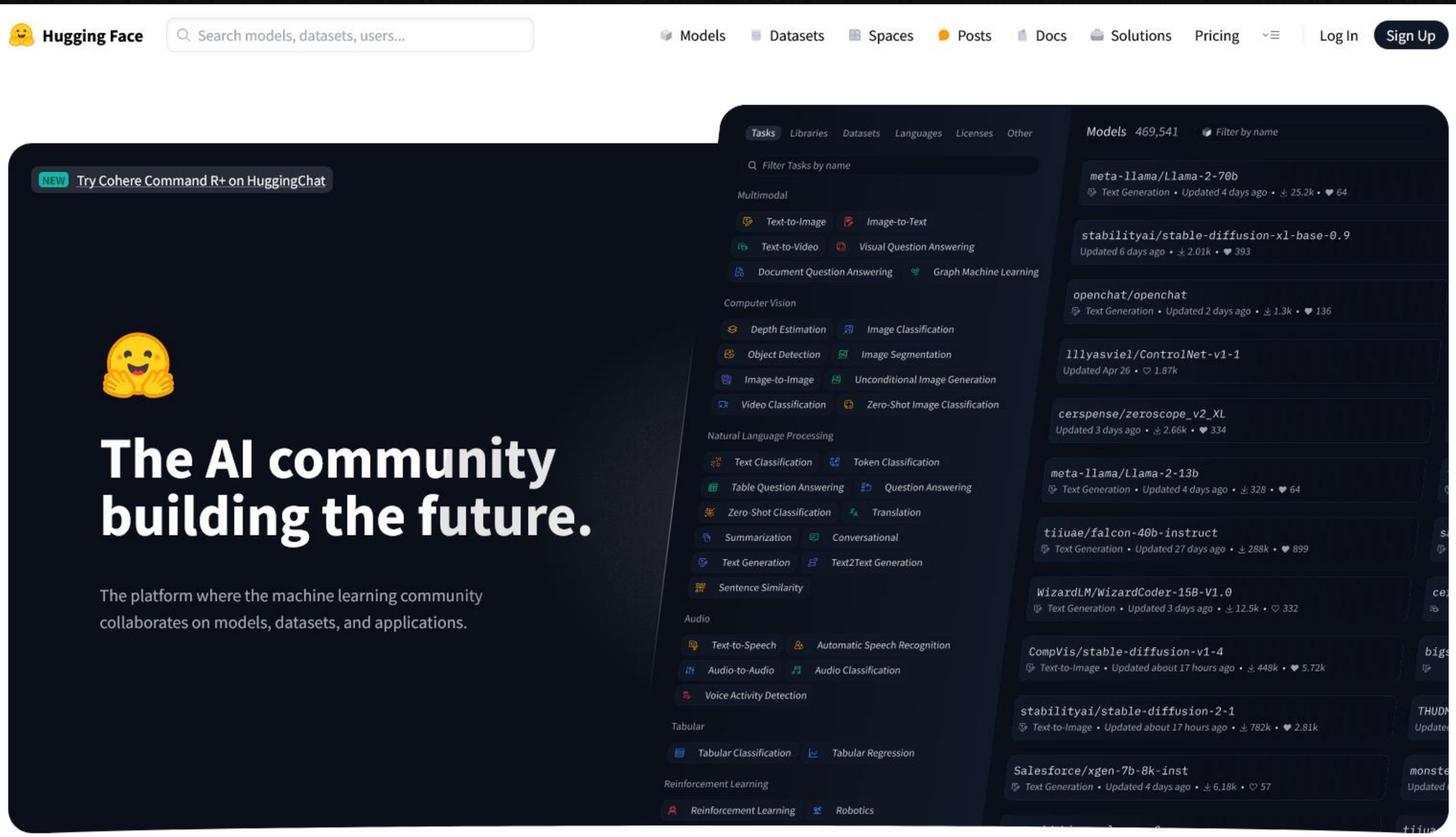


“A BOWL OF SOUP THAT LOOKS LIKE A MONSTER KNITTED OUT OF WOOL.”

“TEDDY BEARS MIXING SPARKLING CHEMICALS AS MAD SCIENTISTS IN A STEAMPUNK STYLE.”

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Zero-Shot Classification
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Text Generation
Text2Text Generation
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Audio-to-Audio
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openchat/openchat
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Text Generation • Updated 3 days ago • 12.5k • 332
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stabilityai/stable-diffusion-2-1
Text-to-Image • Updated about 17 hours ago • 782k • 2.81k
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openai / whisper-large-v2 | like 487

Automatic Speech Recognition | PyTorch | TensorFlow | JAX | Transformers | 99 languages | whisper | audio | hf-asr-leaderboard | arxiv:2212.04356

License: apache-2.0

Model card | Files and versions | Community 42 | Train | Deploy | Use in Transformers

Edit model card

Whisper

Whisper is a pre-trained model for automatic speech recognition (ASR) and speech translation. Trained on 680k hours of labelled data, Whisper models demonstrate a strong ability to generalise to many datasets and domains **without** the need for fine-tuning.

Whisper was proposed in the paper Robust Speech Recognition via Large-Scale Weak Supervision by Alec Radford et al. from OpenAI. The original code repository can be found here.

Compared to the Whisper large model, the large-v2 model is trained for 2.5x more epochs with added regularization for improved performance.

Disclaimer: Content for this model card has partly been written by the Hugging Face team, and parts of it were copied and pasted from the original model card.

Model details

Downloads last month
70,522



Hosted inference API

Automatic Speech Recognition | Examples

Browse for file or Record from browser or

Realtime speech recognition

Librispeech sample 1

0:00 / 0:14

Compute

⚠ This model could not be loaded by the inference API. ⚠

Model openai/whisper-large-v2 does not exist

JSON Output | Maximize